Ali Husain

Software engineer and product designer

I build complete systems, from the interface to the infrastructure.



ali@gallowforge.dev

gallowforge.github.io github.com/gallowforge

Remote, UTC+3 · Open to remote work

IN NUMBERS

11
public repositories, each one specified, tested, documented and released

4,503
automated tests across them, run on every change

15
self-hosted services in production on a platform I designed and operate

SELECTED WORK

Holdfast

COMMAND-LINE TOOL
OPEN SOURCE

A memory for coding agents in which every fact carries its own proof.
Each fact is stored with a check, run again before the agent is told. What no longer holds comes back as broken.

1,651 automated tests, on Linux and macOS
10 kinds of check, from a file to a live endpoint

gallowforge.github.io/holdfast

Seaward

COMMAND-LINE TOOL
OPEN SOURCE

Shows which ports on a server can really be reached from outside, and the rule that decides each one.
Follows a packet through the rules the kernel has loaded, so a port Docker published past the firewall's list is found.

1,212 automated tests, run on every change
2,111 real packets checked against the kernel

gallowforge.github.io/seaward

Ballast

LIVE APP
OPEN SOURCE

Personal finance that turns savings into a salary that lasts.
Designed and built end to end, offline-first. Says what is safe to spend today and reads bank statements on the device.

599 automated tests
40 currencies

gallowforge.github.io/ballast

Ethereal

LIVE APP
OPEN SOURCE

A calm coach for training, food and the shape of the day.
An installable app that plans the day, guides workouts and tracks nutrition. Designed, built and delivered end to end.

397 automated tests
1,474 foods, offline

gallowforge.github.io/Ethereal

Evalbench

SELF-HOSTED APP
OPEN SOURCE

Names the language model to use for your own job, and shows the proof.
Finds the cheapest model that clears your bar. Answers are graded by automatic checks, then rated blind in pairs.

291 automated tests, plus sandbox checks in CI
7 check types, up to sandboxed code tests

gallowforge.github.io/evalbench

Vesper

SELF-HOSTED APP
OPEN SOURCE

A read-only operations dashboard that answers one question: does anything need you?
One Go binary, read-only by construction. When it cannot confirm something it says so instead of showing an all-clear.

7.4 MB one Go binary, about 27 MB of memory
191 tests, 34 end to end under real systemd

gallowforge.github.io/vesper

Homelab

PLATFORM
OPEN SOURCE

Fifteen services on one small machine, run as code, with backups that are proven every week.
Zero port forwards. Every backup set is verified nightly, copied to three machines and restored in a weekly drill.

15 containers, 21 timers, 43 automated tests
77 alert conditions and 13 uptime probes

github.com/gallowforge/homelab

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CAPABILITIES		Where to see it
Product design	Interfaces designed for phone and desktop together, accessible by default, and measured with real devices and automated audits.	Ballast, Ethereal, Vesper Ethereal scores Lighthouse 100 on key pages, swept against WCAG 2.1 AA
Front end	Installable apps that work offline, with no account and no server behind them, and a live operations dashboard streamed to the browser.	Ballast, Ethereal, Vesper 599 and 397 automated tests on the two apps
Back end	Services and command-line tools in Python, TypeScript, Go, Bash, SQL and C/C++, with the documentation that ships with them.	Holdfast, Seaward, Vesper, Evalbench Three Go binaries, each with no third-party Go dependencies
AI evaluation and agent workflows	Model evaluation with automatic checks and blind human rating. Agent memory that is proved again before use, and reviewer and verifier agents that judge work they did not write.	Holdfast, Evalbench, agent-guardrails One build ran as nine coordinated agents, a platform audit as thirteen
Infrastructure and operations	Linux, containers, systemd, CI/CD and infrastructure as code, with monitoring and incident response. Backups are proven by a restore drill every week.	Homelab, Vesper 410 commits, 21 systemd timers, about 4,100 lines of runbooks
Security practice	DNS, TLS, mesh VPN, firewalls, service sandboxing, intrusion prevention, encryption and secrets handling. Zero port forwards and one audited public endpoint.	Seaward, Homelab, Vesper, Evalbench 7-node mesh VPN; code tests run in a sandbox with no network
Video pipeline	A documented production method for painted films made with AI image and video models, with post-production built in code, from beat-matched editing to 4K mastering on GPU.	filmforge 9 skills and 7 tested tools; every encode verified

HOW I WORK

Specified before it is built, tested before it ships, documented so it runs without me.

Prove it works	A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.
Write it down	Every system ships with a runbook someone else could follow.
Learn by shipping	Linux, Go, AI agents, post-production: each learned by building something real.
Fewer things, done properly	Simple setups that stay up beat clever ones that break.

Smaller tools · github.com/gallowforge

agent-guardrails 36 tests
Agents that judge work they did not write, and a warning before a destructive command.

aih 31 tests
Every AI conversation from every machine in one local full-text index.

bwtidy 12 tests
Audits and tidies a password vault, with a plan you approve before anything changes.

filmforge 40 tests
A tested production method for painted films made with AI models.

EDUCATION

Computer Science and Engineering

CONTACT

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